



8-Queens:

Classic algorithms,
Challenges, and
Innovation

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THE 8-QUEENS PROBLEM

Overview

The Problem

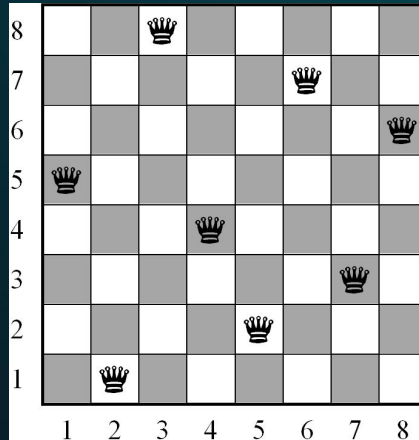
Place 8 queens on a chessboard such that none attack each other.

Conditions

No two queens share the same row, column or diagonal

Generalization

Problem extended to placing n -queens on a $n \times n$ chessboard.



Origin

In 1848 by mathematician Max Bezzel.

Historical Interest

Gauss proposed early methods to solve it.

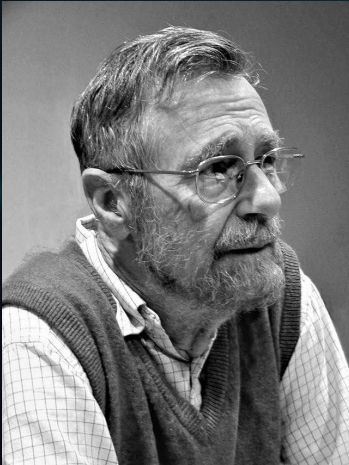
Dijkstra Contribution

Demonstrated structured programming using the n -queens problem.

DIJKSTRA'S CONTRIBUTION

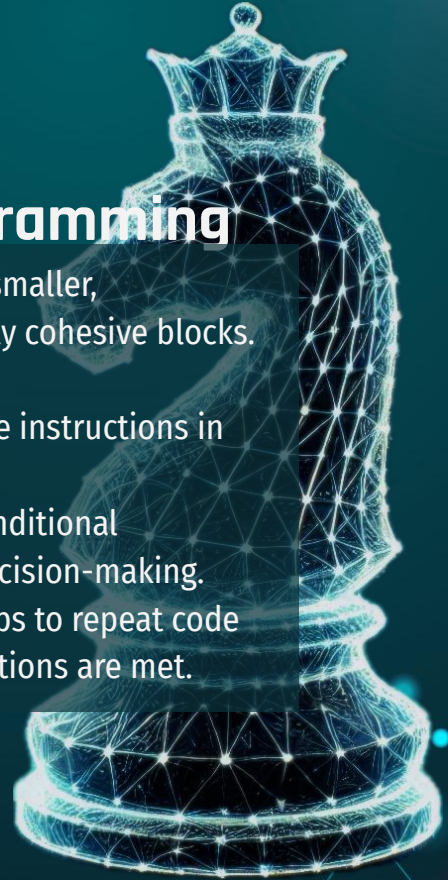
Dijkstra in 1972

Used the n-queens problem to demonstrate the power of structured programming



Structured Programming

- Breaking programs into smaller, manageable, and logically cohesive blocks.
- Core Constructs:
 - **Sequence:** Execute instructions in order.
 - **Selection:** Use conditional statements for decision-making.
 - **Iteration:** Use loops to repeat code blocks until conditions are met.



GENERALIZATION AND MATHEMATICAL SIGNIFICANCE

- **Generalization to the n-Queens:**
 - Problem extended to placing n-queens on a $n \times n$ chessboard.
- **Key statistics for n=8**
 - Total configurations: 4,426,165,368.
 - Valid solutions: 92 (or 12 unique solutions considering rotations and reflections)
- **Combinatorial Insight:**
 - The problem showcases the challenges and elegance of combinatorial optimization and algorithmic design.



COMPARISON OF ALGORITHMS



Hill Climbing



**Simulated
Annealing**

**Classic
Algorithms**



**Genetic
Algorithms**



**Depth-First
Search**

HILL CLIMBING - 8-QUEENS

Process

The algorithm begins with a random placement of 8 queens on the chessboard



Initialization



Successor Generation



Explore all possible neighboring moving queen at a time within its column. ($8 \times 7 = 56$ potential configurations)

Pick the neighbor with the lowest h-value. If tied, chose randomly.



Selection



SIMULATED ANNEALING - 8-QUEENS

WHAT SETS IT APART



Random Moves

Instead of always choosing the best move, worse moves are occasionally accepted



Acceptance Probability

$$e^{-\Delta E/T}$$

Temperature (T)

High temperatures allow greater randomness, while low temperatures focus on refinement.



The "badness" (ΔE)

Larger increases in h are less accepted



GENETIC ALGORITHMS - 8-QUEENS

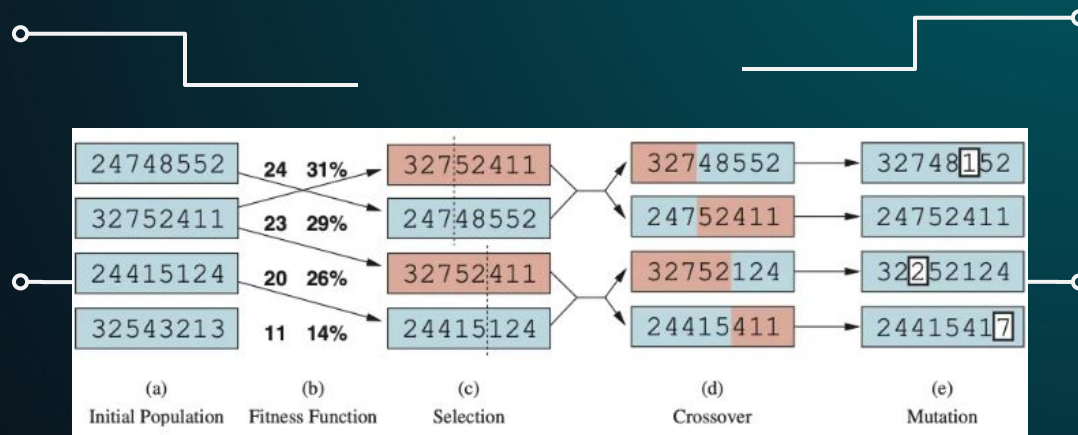
How they work

POPULATION

The string denotes queen positions in each column

FITNESS FUNC

Measures the number of non-attacking pairs of queens



SELECTION

Individuals are chosen based on fitness function

CROSSOVER

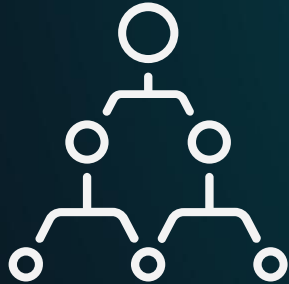
Individuals exchange parts of their structure to create offspring

MUTATION

A small, random change is applied to some offspring

DEPTH-FIRST SEARCH - 8-QUEENS

REPRESENTATION

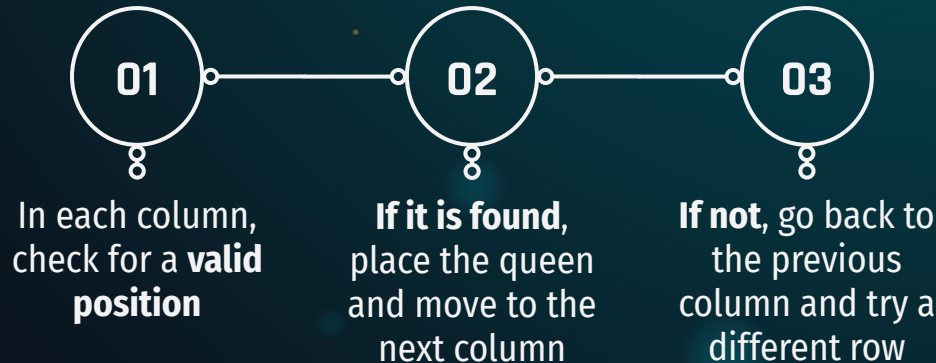


Decision Tree where:

- **Levels** correspond to the columns of the board
- **Nodes** represent possible row placements in each column
- **Paths** indicate complete configurations, from start point to the solution.

PROCESS

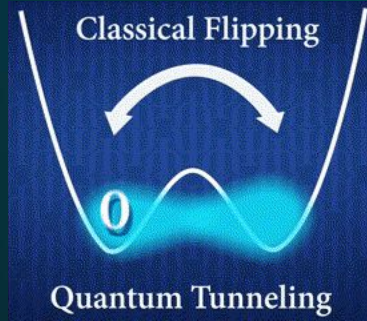
The algorithm begins with an empty board and recursively attempts to place queens.





INNOVATIONS AND PROSPECTS

QUANTUM ANNEALING - 8-QUEENS



QUANTUM

ALGORITHMS

By leveraging the unique properties of **quantum mechanics**, these algorithms explore solution spaces in a different way

SUPERPOSITION

Particle can exist in multiple states simultaneously

QUANTUM

TUNNELLING

Particle passes through a barrier that it normally shouldn't be able to cross.

PARALLEL

EXPLORATION

Multiple board arrangements are explored in parallel.

AVOIDING LOCAL

OPTIMA

Bypassing high-energy barriers, the system escapes local traps.

SWARM INTELLIGENCE ALGORITHMS

Collective intelligence and its impact on metaheuristics



FORAGING

Ants lay down pheromone trails while searching for food

FOLLOWING

Other ants follow trails based on pheromone concentration



FEEDBACK

Successful paths are reinforced through higher pheromone levels

OPTIMIZATION

The colony converges on the shortest or most efficient path



SWARM INTELLIGENCE ALGORITHMS



Swarm Intelligence is inspired by collective behaviors in nature, like ants or bees



Ant Colony Optimization mimics ants' foraging, where pheromone trails guide others to the best paths.



Particle Swarm Optimization mimics bird flocking, with particles exploring solutions based on personal and swarm bests.



Artificial ants explore solutions, leaving pheromones that evolve to lead to optimal results.



Through iterative updates, it's balanced exploration and exploitation to find the optimal solution.

PARTICLE SWARM OPTIMIZATION

8-QUEENS PROBLEM

REPRESENTATION

Each particle is an array of 8 numbers, representing queen positions by column per row

FITNESS

Measures the number of conflicts in a configuration

POSITION

A particle's position updates by adding its velocity.

HOW IT WORKS



PARTICLE

Each particle has a position and velocity

VELOCITY

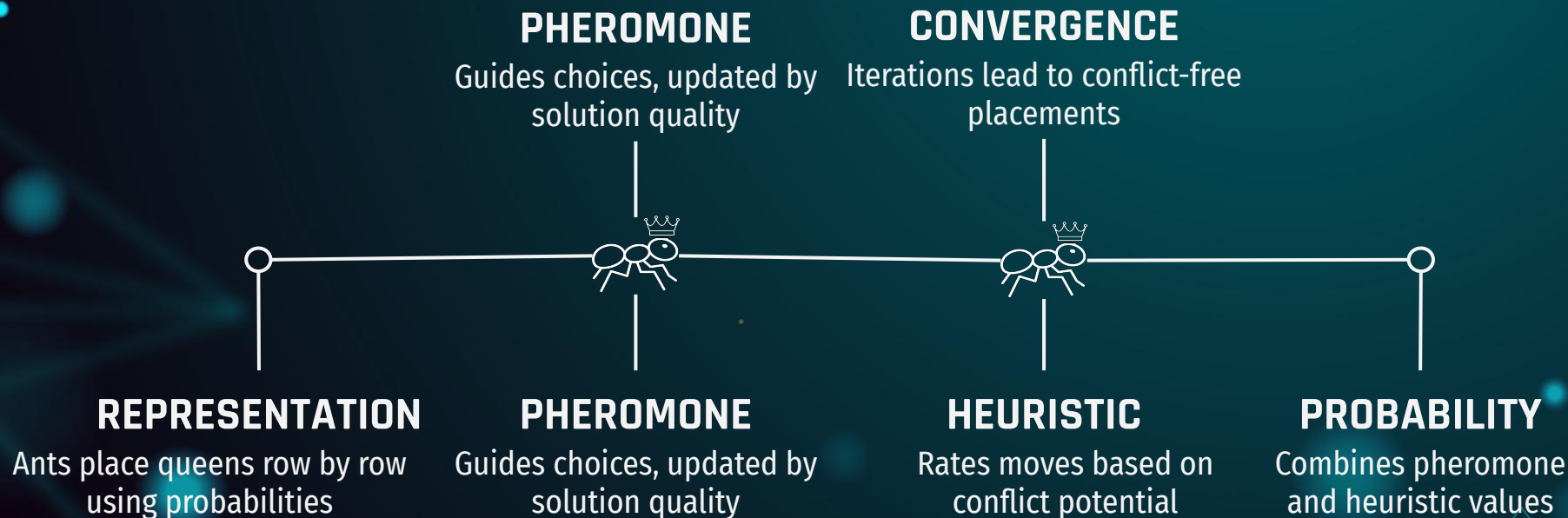
Velocity depends on the particle's and swarm's best solution; inertia, and acceleration, for exploration and exploitation

CONVERGENCE

Iterative updates minimize conflicts for valid solutions

ANT COLONY OPTIMIZATION

8-QUEENS PROBLEM



REAL-WORLD PROBLEMS

INSPIRED BY THE 8-QUEENS PROBLEM



BIOINFORMATIC

Problems like sequence alignment or protein structure modeling use similar techniques to avoid invalid configurations



LOGISTIC

Optimizing routes for drones, vehicles, or aircraft to avoid collisions mirrors the queen placement strategy to prevent conflicts

OPEN CHALLENGES

FUTURE DIRECTIONS

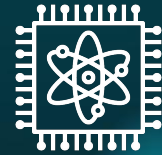
COMPLEXITY

Extensions of the
n-queens problem with
additional constraints



INNOVATIONS

Impact of quantum computing
and supercomputers on
combinatorial optimization
problems



THANKS FOR YOUR ATTENTION!